# Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

**SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT** **JEE-MAIN** **Date: 11-01-2026**
**Time: 02.00Pm to 05.00Pm** **GTM-12** **Max. Marks: 300**

## KEY SHEET

## MATHEMATICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **3** | 2 | **1** | 3 | **2** | 4 | **1** | 5 | **3** |
| 6 | **2** | 7 | **1** | 8 | **2** | 9 | **4** | 10 | **1** |
| 11 | **3** | 12 | **3** | 13 | **4** | 14 | **1** | 15 | **2** |
| 16 | **4** | 17 | **4** | 18 | **4** | 19 | **1** | 20 | **1** |
| 21 | **3** | 22 | **0** | 23 | **2** | 24 | **0** | 25 | **4** |

## PHYSICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 26 | **4** | 27 | **3** | 28 | **3** | 29 | **3** | 30 | **1** |
| 31 | **3** | 32 | **3** | 33 | **3** | 34 | **4** | 35 | **4** |
| 36 | **3** | 37 | **3** | 38 | **1** | 39 | **3** | 40 | **2** |
| 41 | **2** | 42 | **2** | 43 | **3** | 44 | **1** | 45 | **3** |
| 46 | **25** | 47 | **15** | 48 | **2** | 49 | **343** | 50 | **4** |

## CHEMISTRY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 51 | **4** | 52 | **3** | 53 | **2** | 54 | **2** | 55 | **4** |
| 56 | **1** | 57 | **1** | 58 | **4** | 59 | **2** | 60 | **1** |
| 61 | **3** | 62 | **1** | 63 | **2** | 64 | **1** | 65 | **4** |
| 66 | **3** | 67 | **1** | 68 | **3** | 69 | **4** | 70 | **3** |
| 71 | **5** | 72 | **1** | 73 | **63** | 74 | **4** | 75 | **2** |



For More Join: @IITJEE_Advanced

# SOLUTION

# MATHEMATICS

1. $S=\frac{1}{3}\left[\log\frac{1}{2}+\log\frac{2}{3}+\log\frac{3}{4}+.....+\log\frac{99}{100}\right]=\frac{1}{3}\left[\log\frac{1}{2}.\frac{2}{3}.\frac{3}{4}.....\frac{99}{100}\right]$ $S=\frac{1}{3}\log_{10}\frac{1}{100}=\frac{-2}{3}$

2. $\alpha+\beta+\gamma=0 \text{ and } \alpha\beta\gamma=-10$

   Given expression $=-\gamma^3-\alpha^3-\beta^3$ $=\left(\alpha^3+\beta^3+\gamma^3\right)=-3\alpha\beta\gamma=30$

3. $A.M\geq H.M \Rightarrow \frac{\frac{3}{\alpha+1}+\frac{3}{\beta+1}+\frac{3}{\gamma+1}}{3}\geq\frac{3}{\frac{\alpha+1}{3}+\frac{\beta+1}{3}+\frac{\gamma+1}{3}}$

   $\frac{3}{\alpha+1+\beta+1+\gamma+1}\geq\frac{3.3.3}{(\alpha+\beta+\gamma)}\geq\frac{27}{9}(\alpha+\beta+\gamma=6)$ $\Rightarrow\frac{3}{\alpha+1}+\frac{3}{\beta+1}+\frac{3}{\gamma+1}\geq 3$

4. $1B\,2B\,3B\,4B\,5,4!\times(123 \text{ or } 234 \text{ or } 345)=4!.3!$

5. We have $(x+2)(x+3)(x+4).....(x+10)\equiv a_9x^9+a_8x^8+....+a_1x+a_0....(1)$

   Put $x=1$ $a_9+a_8+a_7+.....+a_1+a_0=\frac{11!}{2}$

   $\underset{(\text{Adding})}{x=-1}\frac{a_9-a_8+a_7+....+a_1-a_0=-9!}{}$ $\Rightarrow 2(a_9+a_7+....+a_1)=\frac{11!}{2}-9!=(55-1)9!$

   So, $a_9+a_7+....+a_1=27(9!)$

6. $\begin{vmatrix}0 & a-c & b-c\\ 0 & c-b & a-c\\ 1 & b & c\end{vmatrix}=0$ $\therefore\Rightarrow A=B=C=60^0\Rightarrow\sin^2A+\cos^2B+\tan^2C$ $=\frac{3}{4}+\frac{1}{4}+3=4$

7. $n(S)=BGG(3);\ BBG(3);\ BBB(1);$ hence $n(S)=7$ $n(A)=BBG(3)$ $\Rightarrow P=\frac{3}{7}$

8. $x+iy=\lambda+3-i\sqrt{5-\lambda^2}$ $x=\lambda+3, y=-\sqrt{5-\lambda^2}$

9. $y=\cos x(2\sin 2x.\sin x)=\sin^2 2x$ $\therefore y\geq 0\forall x\in R$

   Aliter: $2y=2\cos^2x-2\cos x\cos 3x=(1-\cos 2x)-(\cos 4x+\cos 2x)$

   $=1-\cos 4x=2\sin^2 2x$ $y=\sin^2 2x\geq 0\forall x\in R$

10. In the right angled triangle circum centre is midpoint of hypotenuse

11. Let the equation of the circle be $x^2+y^2+2gx+2fy+c=0$

    Since it passes through $(1,2)$, so

    $1+4+2g+4f+c=0$ $\Rightarrow 2g+4f+c+5=0$

12. $SP_1=a\left(1+t_1^2\right);$ $SP_2=a\left(1+t_2^2\right)$

    Also $t_1t_2=-1$ $\therefore\frac{1}{SP_1}+\frac{1}{SP_2}=\frac{1}{a}$

13. Since $1^2+2^2-9<0$ and $2^2+1^2-9<0,$ both P and Q lie inside C.

    Also, $\frac{1^2}{9}+\frac{2^2}{4}-1>0 \text{ and } \frac{2^2}{9}+\frac{1^2}{4}-1<0, P$ lies outside E and Q lies inside E.

    Thus, P lies inside C but outside E.

14. Chord of the hyperbola $x^2-y^2=a^2$ with middle point $(h,k)$, is



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$hx-ky=h^2-k^2 \quad \Rightarrow y=\frac{h}{k}x+\frac{k^2-h^2}{k}.....(1)$

As $(1)$ touches $y^2=4ax,$ condition of tangency gives

$c=\frac{a}{m}\Rightarrow\frac{k^2-h^2}{k}=\frac{ak}{h}$

15. Let $\vec{d}=x\hat{i}+y\hat{j}+z\hat{k}$ and $\left[\vec{b}\ \vec{c}\ \vec{d}\right]=0$ gives $x+y+z=0$

$x=y\ and\ z=-2x\ \therefore \vec{d}=\pm\frac{\hat{i}+\hat{j}-2\hat{k}}{\sqrt{6}}$

16. It's direction ratiois <3, 1, 0>

$\therefore$ option(a) and (b) are correct.

Any point on the line is $<3+3\lambda,2+\lambda,1>$

Hence, for $\lambda=1,$ option (c ) is correct

Line and plane are parallel to each other.

Hence , option (d) is incorrect

17. Plane P passes trough $(1,5,3)$ and normal to $\vec{A}=(2,3,6)$

Equation is $2(x-1)+3(y-5)+6(z-3)=0$

$\Rightarrow 2x+3y+6x=35 \quad \therefore$Perpendicular from $(0,0,0)$ to the plane $=\left|\frac{-35}{7}\right|=5$

18. Let $y=\frac{x^2+2x+a}{x^2+4x+3a}\Rightarrow(y-1)x^2+(4y-2)x+(3ay-a)=0$

Since $x\in R,\ \Delta\geq 0 \quad \Rightarrow(4-3a)y^2-4(1-a)y+(1-a)\geq 0,\forall y\in R$

$\Rightarrow 4-3a>0\ and\ 16(1-a)^2-4(4-3a)(1-a)\leq 0 \quad \Rightarrow 0<a<1$

19. $\tan^{-1}\frac{1}{\sqrt{2}}-\left(\tan^{-1}\sqrt{3}-\tan^{-1}\sqrt{2}\right)=\frac{\pi}{6}$

20. If $|x|\leq l\Rightarrow -l\leq x\leq l.$

21. $\sin\theta.\sin\left(60^0-\theta\right)\sin\left(60^0+\theta\right)=\frac{1}{4}\sin 3\theta$

22. $\int_{-a}^{a}(\text{odd function})\,dx=0$

23. $f'(x)=3x(x-2)$

Critical points $x=0,2$

24. $f(x)=\frac{x^3}{3}-(m-3)\frac{x^2}{2}+mx-2013$

$\therefore f'(x)=\left(x^2-(m-3)x+m\right)\geq 0,\forall x\in[0,\infty)$

Case I: When $D\leq 0\Rightarrow m\in[1,9]$

Case II: When $D\geq 0\Rightarrow m\in(-\infty,1]\cup[9,\infty)$

$\frac{-b}{2a}\leq 0\Rightarrow m\leq 3 \qquad f'(0)\geq 0\Rightarrow\ m\geq 0$

25. Area $=\int_{-2}^{2}(2-|x|)\,dx \quad =\int_{0}^{2}(2-x)\,dx$



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# PHYSICS

26. Use $\alpha=\beta=45^0$ in the formula for Range down the incline plane.

27. $\vec{V}_{m.g}=\vec{V}_{m,r}+\vec{V}_{r,g}$

As resulting velocity $\vec{V}_{m.g}$ is at $45^0$ with river flow

$i.e=\vec{V}_{r,g}-\vec{V}_{m,r}\sin\alpha=V_{m,r}\cos\alpha$ .....(1)

And $\frac{60m}{V_{mr}\cos\alpha}=6\sec$ .....(2)

Solving (1) & (2)

$V_{m,r}=5m/s$

28. For block to be stationary $T=800N$

If man moves up by acc.'a'

500 N

$T-mg=ma$

$800-500=50a$

$a=6m/s^2$

29. Limiting condition for m to not slip in vertical downward direction, $mg=\mu N$

$\Rightarrow N=\frac{mg}{\mu}=\frac{100}{.5}=200\,Newton$

Same normal force would accelerate M, thus $a_M=\frac{200}{50}=4m/s^2$

Taking $m+M$ as system

$F=(m+M).4=240N$

30. As long as the block of mass m remain stationary, the block of mass M released from rest comes down by $\frac{2Mg}{K}$ (before coming it rest momentanly again).

Thus the maximum extension in spring is

$x=\frac{2Mg}{K}$.....(1)

For block of mass m to just move up the incline

$kx=mg\sin\theta+\mu mg\cos\theta$.....(2)

$2Mg=mg\times\frac{3}{5}+\frac{3}{4}mg\times\frac{4}{5}or\ M=\frac{3}{5}m$

31. $X^2=4ay$

Differentiating w.r.t.y, we get

$\frac{dy}{dx}=\frac{x}{2a}$

$\therefore \text{At}(2a,a),\frac{dy}{dx}=1\Rightarrow$ hence $\theta=45^0$

The component of weight along tangential direction is $mg\sin\theta$.



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Hence tangential acceleration is $g\sin\theta=\frac{g}{\sqrt{2}}$

32. Let $v_1, v_2$ and $v_3$ be velocities of blocks 1,2 and 3 after suffering collision each.

$mv=mv_1+mv_2$ and $v_1$-$v_2$=-v

Solving we get $v_1=\frac{m-M}{m+M}$

Similarly $v_3=\frac{2m}{m+M}\times v_2=\frac{4Mmv}{(m+M)^2}.....(2)$

$\therefore \frac{M-m}{M+m}v=\frac{4Mmv}{(M+m)^2}$ $or\, M^2-m^2=4Mm$ $\frac{M}{m}=2+\sqrt{5}$

33. Statement- 2 Contradicts Newton's third law and hence is false.

34. As the disc is in combined rotation and translation, each point has a tangential velocity and a linear velocity in the forward direction.

From figure

$V_{net}$ (for lowest point) = v-R$\omega$=v-v=0

And Acceleration = $\frac{V^2}{R}+0=\frac{V^2}{R}$

Since linear speed is constant) Hence (D)

35. For a disc rolling without slipping on horizontal rough surface with uniform angular velocity, the acceleration of lowest point of disc is directed vertically upwards and is not zero. Hence statement 1 is false.

36. Both the spring are in series

$\therefore K_{eq}=\frac{K(2K)}{K+2K}=\frac{2K}{3}$

Time period $T=2\pi\sqrt{\frac{\mu}{K_{eq}}}$ where $\mu=\frac{m_1m_2}{m_1+m_2}$

Here $\mu=\frac{m}{2}$ $\therefore T=2\pi\sqrt{\frac{m}{2}.\frac{3}{2K}}$ $=2\pi\sqrt{\frac{3m}{4K}}$

37. Velocity of sound is inversely proportional to the square root of density of the medium

i.e $V\sqrt{\rho}=$ constant $\Rightarrow \frac{V_1}{V_2}=\sqrt{\frac{\rho_2}{\rho_1}}=\sqrt{\frac{2\rho}{\rho}}=\sqrt{2}$

38. For refraction medium changes so frequency does not change because frequency depends on sources for reflection medium does not change so speed of wave and wave length does not change

Refraction from rarer to denser medium wave length decreases because density of denser medium high reflection from denser medium speed and frequency does not change because medium is same but due to reflection change of $\pi$. Takes place

39. $Q_{AB}=\Delta U_{AB}+W_{AB}$

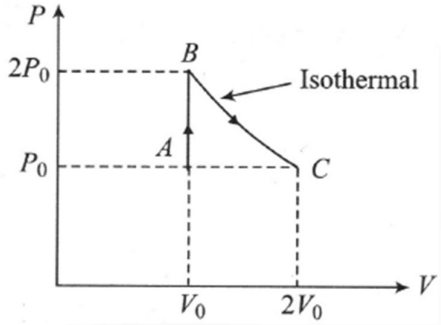


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$W_{AB}=0$

$\Delta W_{AB}=\frac{f}{2}nR\Delta T \qquad \Rightarrow \frac{f}{2}(\Delta PV)\ \Delta U_{AB}=\frac{5}{2}(\Delta PV)$

$Q_{AB}=2.5\ P_0V_0$

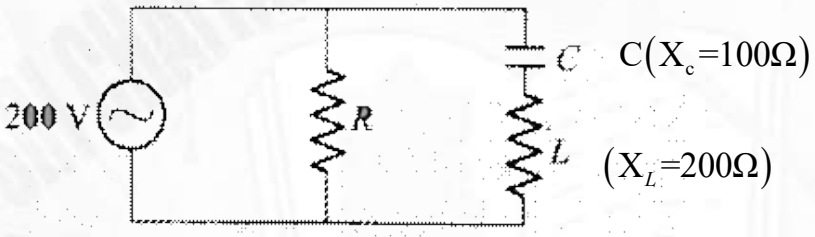



40. $V=\frac{K\vec{p}.\vec{r}}{r^3}=\frac{9\times10^9\times\left(2\hat{i}-3\hat{j}+4\hat{k}\right).\left(2\hat{i}+2\hat{j}-\hat{k}\right)}{\left[2^2+2^2+1^2\right]^{3/2}}=\frac{9\times10^9\times(4-6-4)}{27}$

$V=-2\times10^9\ volts$

41. Both statements 1 and 2 are true. In statement-1 R is varied while in statement -2 R is kept constant.
Hence both statements are independent.

42.




$I_R=\frac{V}{R}=\frac{200}{100}=2A$

$r=\frac{V}{X_L-X_C}=\frac{200}{100}=2A$

$I=\sqrt{I_R^2+I'^2}=2\sqrt{2}\ \text{Amp.}$

43.

At distance X consider small element of width dx

Magnetic moment of the small elements is : $dm=\frac{\left(\frac{q}{l}dx\right)\omega}{2\pi}.\pi x^2$



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$$M=\int_{-l/2}^{l/2}\frac{q\omega}{2l}\times^2 dx,\quad M=\frac{q\omega l^2}{24}=\frac{q\pi fl^2}{12}$$

44. The graph of current is given by:

$$i=i_0\left(1-e^{-t/\tau}\right)\Rightarrow\frac{di}{dt}=\frac{i_0}{\tau}e^{-t/\tau}$$

Energy stored in the form of magnetic field energy is:

$$U_B=\frac{1}{2}Li^2$$

$\therefore$ Rate of increases of magnetic field energy is:

$$R=\frac{dU_B}{dt}=Li\frac{di}{dt}=\frac{Li_0^2}{\tau}\left(1-e^{-t/\tau}\right)e^{-t/\tau}$$

This will be maximum when $\frac{dR}{dt}=0$

$$\Rightarrow R_{\max}=\frac{Li_0^2}{\tau}\left(1-\frac{1}{2}\right)\left(\frac{1}{2}\right)=\frac{Li_0^2}{4\tau}=\left[\frac{L\left(E/R\right)^2}{4\left(L/R\right)}\right]=\frac{E^2}{4R}$$

45. If maximum intensity is observed at P then for maximum intensity to be also observed at $Q, S_1$ and $S_2$ must have phase difference of $2m\pi$ (where m is an integer)

46. Thickness of annular space $=\frac{20.628-20}{2}=.314cm$

In steady state, gravitational force =viscous force

$$\mathrm{M\,mg}=\eta\mathrm{A}\frac{\mathrm{v}}{\mathrm{y}}\qquad 1\times10=10\times10^{-1}\times2\pi rl\frac{v}{y}$$

$$1\times10=10\times10^{-1}\times2\times3.14\times10\times10^{-2}\times20\times10^{-2}\frac{v}{.314\times10^{-2}}\quad 1=4\mathrm{v}$$

$$\mathrm{v}=\frac{1}{4}=0.25m/\sec=25cm/\sec$$

47. By snell's law

$$1\times\sin30^0=.....=.....=n\sin i$$

Where n is R. I. at y and i is angle of incidence at y.

$$\tan\left(90-i\right)=\frac{dy}{dx}=8\mathrm{x}=4\sqrt{y}$$

$$\cot i=4\sqrt{y}=4\sqrt{\frac{1}{2}}=2\sqrt{2}\Rightarrow\sin i=\frac{1}{3}\quad\therefore n=\frac{\sin30^0}{\sin i}=\frac{1/2}{1/3}=1.5$$

48. $\mathrm{KE}_{\max}=\left(5\text{-f}\right)\mathrm{eV}$

When these electrons are accelerated through 5V,

They will reach the anode with energy $=\left(5-\phi+5\right)\mathrm{eV}\quad\therefore10-\phi=8\quad\phi=2eV$

Current is less than saturation current Ans

Because if slowest electron also reached the plane it would have $5eV$ energy at the anode, but there it is given that the minimum energy is $6eV$.

49. $N^{14}+\alpha\rightarrow O^{17}+\text{proton}$



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Q value= (14.00307+4.00260-1.00783-16.99913) 931.5=-1.20 MeV

Let m and M be mass of $\alpha$ particle and nitrogen nucleus respectively and let minimum KE of $\alpha$ particle be $\frac{1}{2}mu^2$.

From energy equation

$$\frac{1}{2}mu^2 = |Q| + \text{minimum KE of system} = |Q| + \frac{1}{2}(m+M)\left[\frac{mu}{(m+M)}\right]^2$$

$$\therefore \frac{1}{2}mu^2\left(\frac{M}{m+M}\right)|Q| \text{ or } \frac{1}{2}mu^2|Q|\left(\frac{m+M}{M}\right)$$

$$\text{KE of products} = \frac{1}{2}(m+M)\left[\frac{mu}{(m+M)}\right]^2 = \frac{1}{2}mu^2\left(\frac{m}{m+M}\right) = |Q|\frac{m}{M}$$

$$= 1.2 \times \frac{4}{14} = 0.3428\,MeV = 342.8\text{KeV}$$

50. The charge on the capacitor when current reaches $I_0$

When the switch is in position 2, this charge is shared with capacitor $C_2$ and at steady state potential across

$C_1$ is equal to that across $C_2$. The energy lost in this process

$$\Delta U = \frac{1}{2}\cdot\frac{C_1C_2}{C_1+C_2}\times\left(\frac{q_0}{C_1}\right)^2 = \frac{1}{2}\cdot\frac{C_1C_2}{C_1+C_2}\times(I_0R)^2 = 4J$$



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# CHEMISTRY

51. $t_1 = \left(\frac{1}{2}\right)^2$ 2 half life time, $t_2 = \left(\frac{1}{2}\right)^4$ 4 half life time

52. z=55 and z= 19 are alkali metal among them I.E least for 55.

53. Phenol and aniline show negative deviation, so boiling point increases.

54. $1s^2 \quad 2s^2 \quad 2p^5$

55. $SF_6$

$(CoF_6)^{3-} \quad (MnCl_6)^{3-}$

56. $NaHCO_3$ - Catboxylic acid

$FeCl_3$ - Phenolic

$CAN$ - alcohol

$OH^- / KMnO_4$ - double bond

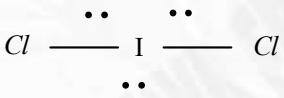
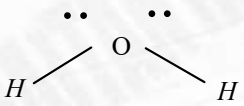
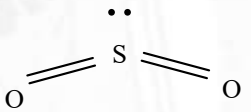
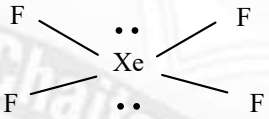
57. $1^0$ amine and aniline

Show carbylamine test

58. 708 Sn $Sn^{+2} \rightarrow R.A$

715 $Pb$ $Pb^{+4} \leftarrow O.A$

59.

| | | BP | LP |
|---|---|---|---|
| $ICl_2^-$ |  | 2 | 3 |
| $H_2O$ |  | 2 | 2 |
| $SO_2$ |  | 2 | 1 |
| $XeF_4$ |  | 4 | 2 |

60.

| | | | |
|---|---|---|---|
| $V^{+2}$ | $4s^0$ | $3d^3$ | $n=3$ |
| $Co^{+2}$ | $4s^0$ | $3d^7$ | $n=3$ |

61. Statement- I false

Alkylhalides undergo elimination to from alkenes

Statement- 2 True

62. Statement – 1

N-N Single bond s weaker

But it is also shorter

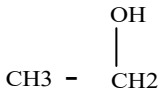
Statement – II

All compounds in +3 oxidation state do not undergo disproportionation

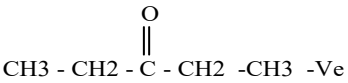


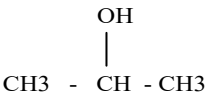
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63.

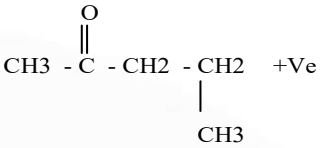


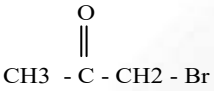

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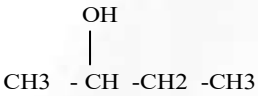




+ ve

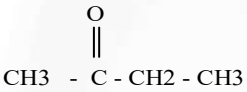


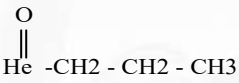

+ ve




+ ve




+ ve

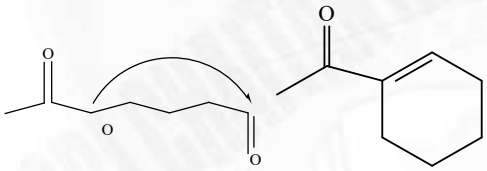



- ve

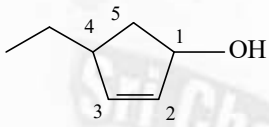
Pentanal -Ve

Pentan - 3- 01 – Ve

64.




65.




4-ethylecopent2-ln-01

66. acidic strength

Statement

I False $H_2O < H_2S < H_2Se < H_2Te$

H- Se

H- Se H- Te

Statement- II Strong bond Weak

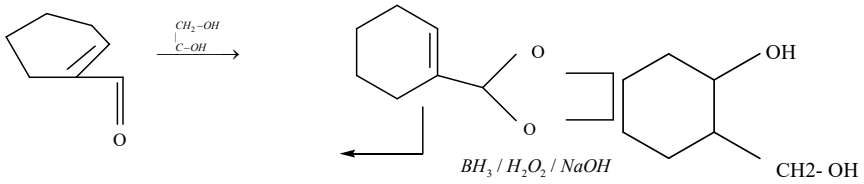
True

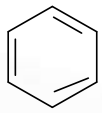
67.

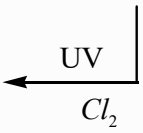
| $SO_2$ | $NO_2^-$ | $N_3^-$ |
|---|---|---|
| $Sp^2$ | $Sp^2$ | $Sp$ |

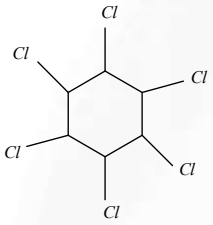


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68. 


69. $C_6H_{12}O_6 \xrightarrow{HI}$ n hexane $\xrightarrow{Cr_2O_3}$ 




70. $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow$

$8MnO_4^- + 6SO_4^{2-} + 2OH^-$

71.

| | | | |
|---|---|---|---|
| $I_3^+$ | $Sp^3$ | $Ni(Co)_4$ | $Sp^3$ |
| $SiO_4^{4-}$ | $Sp^3$ | $XeO_2F_2$ | $Sp^3d$ |
| $SO_2Cl_2$ | $Sp^3$ | $(PtCl_4)^{2-}$ | $dsp^2$ |
| $XeF_2$ | $Sp^3d$ | $XeF_4$ | $Sp^3d^2$ |
| $ClF_3$ | $Sp^3d$ | | |

72. $NaI + AgNO_3 \rightarrow AgI + NaNo_3$

$10^{-3} \times 20 \times M = \frac{4.74}{235}$ $M = \frac{0.02}{20} \times 10^3$ $M = \frac{20}{20} = 1$

73. $CaCO_3 \rightarrow CaO + CO_2$

$100g \rightarrow 56g$ $150 \times 0.75$ $x = \frac{156 \times 0.75 \times 56}{100}$ $= 63kg$

74. All of them show tautomerism

75.

| | | |
|---|---|---|
| $(Co(NH_3)_6)^{+3}$ | | Dia |
| $(Co(C_2O_4)_3)^{3-}$ | | Dia |
| | | $Mn^{+3}$ $4s^0$ $3d^4$ |
| $(MnCl_6)^{3-}$ | Para | n=4 |
| $(Mn(CN)_6)^{3-}$ | Para | n=3 |
| $(CoF_6)^{3-}$ | Para | n= 4 |
| $(Fe(CN)_6)^{3-}$ | para | n= 1 |
| $(FeF_6)^{3-}$ | Para | n=5 |



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